

Mohamed Shakeel

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Summary

Generative AI Developer and AI/ML Engineer – Computer Science graduate (2026) who builds AI systems end to end — training ML models, integrating LLMs, and applying prompt engineering in production. Proficient in Python, FastAPI, and Scikit-Learn, with deployed work spanning intelligent APIs, RAG pipelines, and full-stack web and mobile AI products.

Education

Nellai College of Engineering, Tirunelveli, India	2022 – 2026
<i>Bachelor of Engineering – Computer Science and Engineering</i>	<i>CGPA: 7.83</i>
Little Flower Matriculation Higher Secondary School, Tirunelveli, India	2021 – 2022
<i>Higher Secondary Certificate (HSC)</i>	<i>Percentage: 86%</i>

Skills

AI and Machine Learning: Large Language Models, Retrieval-Augmented Generation, Model Context Protocol (MCP), Prompt Engineering, LLM API Integration (Groq, Llama 3), Natural Language Processing (NLP), Model Serving and Deployment, Classification Models, Regression Models, Gradient Boosting, Random Forest, XGBoost

Coding Agents and Harnesses: OpenCode, Antigravity, GitHub Copilot, Cursor, Hermes, Pi, Deepseek Harness

Languages: Python, Dart, TypeScript, JavaScript, Java, C

Frameworks and Libraries: FastAPI, Express.js, Pydantic, Streamlit, Pandas, NumPy, Scikit-Learn, LangChain, LangGraph, Next.js 14, React.js, Node.js, Vite, Tailwind CSS, shadcn/ui, GSAP, Framer Motion, Three.js, Flutter

Databases and Infrastructure: PostgreSQL, Supabase, MongoDB (Motor, Beanie), Redis (Queues, Caching), MinIO, Amazon S3, Docker Compose

Architecture and Security: Microservices, Monorepo (npm Workspaces), Clean Architecture, REST APIs, JWT/OAuth2 Authentication, bcrypt, Rate Limiting, Zod

Mobile: Flutter, Dart, Riverpod, GoRouter, Firebase Auth, Firestore, SQLite, Local Notifications, PDF Generation, Health Data Integration (HealthKit, Google Fit), Android Studio

Tools and DevOps: Docker, Git, GitHub Actions (CI/CD), Nginx, Vitest, SuperTest, Load Testing, node-cron, n8n, Vercel, Render, Railway

Data Visualization: Chart.js, Recharts, FL Chart

Experience

Phoenix Softech	Jul 2025 – Aug 2025
<i>Full Stack Developer Intern</i>	<i>Madurai, India (Remote)</i>

- Developed and deployed Java-based full stack applications, strengthening understanding of software development lifecycle and object-oriented programming principles.
- Debugged and resolved software defects, improving system reliability and application performance by identifying root causes and applying targeted fixes.
- Documented technical processes, API workflows, and solutions to support team knowledge sharing and onboarding.

Projects

CardioGuard – Cardiovascular Disease Risk Prediction System

2025 – 2026

Team Lead / Machine Learning Engineer

- Led a 4-member team to build an end-to-end cardiovascular disease risk prediction system, trained a Gradient Boosting Classifier on 68,645 patient records achieving ROC-AUC of 0.8017 and 73.4% accuracy via 5-fold stratified cross-validation, outperforming Random Forest, Logistic Regression, and XGBoost baselines.
- Engineered 4 derived features – BMI, pulse pressure, mean arterial pressure, and age in years – from raw biometric inputs; performed outlier removal on 1,355 medically impossible records before training to improve model integrity.
- Built a FastAPI backend with a /predict endpoint computing derived features at inference time and returning risk level (Low, Medium, High) with a probability score; integrated with a Flutter mobile app and Next.js 14 web app via Supabase PostgreSQL.

Ayra Saloon – Multi-Slot Booking System

2026 – Present

Freelance Full Stack Developer (Team)

- Built a responsive customer web booking platform in React 18, Vite, and Tailwind CSS for a unisex salon client, architecting a 6-step wizard with gender-filtered service catalogs, duration-based back-to-back slot calculations with 30-min grace overflow, per-service stylist routing, and cascade consecutive-slot suggestions with strict contiguity and back-navigation conflict resolution.
- Engineered a stateless FastAPI and MongoDB backend with atomic delete-on-free slot models and compound unique indexes for race protection; developed a 5-state booking lifecycle engine supporting two-way reschedule proposals via customer self-service portals, 24h TTL auto-decline, and automated RFC 5545 iCalendar (.ics) synchronization.
- Developed a companion operations suite featuring a cross-platform Flutter admin app and split React web panel for staff queue triage, in-salon walk-in bookings with 10-min cutoff overrides, and stylist time-off management with 409 conflict checks; delivered via staged PRs with CODEOWNERS and a safe-base rollback strategy; currently in the final testing stage ahead of client handover.

Anna University Results Portal – Distributed Systems PoC

2026

System Architect

- Architected a queue-based distributed system to handle 1.5 million concurrent users during Anna University Results Day, replacing a crash-prone live SQL architecture with a scalable static-first model.
- Implemented a Redis-backed virtual waiting room with Proof-of-Work CAPTCHA and pre-generated 1.5 million static JSON result files served via MinIO and Amazon S3, fully containerized with Docker Compose.
- Designed the system entirely as a proof-of-concept to demonstrate that infrastructure-level decisions, not application code, are the primary lever for handling extreme traffic at zero marginal cost per additional user.

Stride – AI-Powered Flutter Health Platform

2026

Flutter Developer

- Integrated Groq LLM (Llama 3.3 70B) as a clinical AI assistant with context-aware health insights derived from real-time biometric data, offline fallback heuristics, and an executive PDF report generator with 24-hour bar charts and direct WhatsApp sharing.
- Built a production-grade Flutter health tracking app integrating the device hardware pedometer chip for native step tracking with a real-time sensor engine computing ACSM MET calories, cadence, and pace; architected with Riverpod, GoRouter, SQLite, and Firestore.
- Implemented per-user multi-account data isolation with Firestore cloud sync, offline-first SQLite step delta merging, and automatic conflict resolution ensuring zero data loss when the app resumes from background.

Certifications

Introduction to Generative AI (MDL-388), IBM SkillsBuild

Aug 2025

Micro Certification – Now Platform, ServiceNow

Sep 2025

Foundations of Cybersecurity, Google

May 2024